SOILS4MED project

SOIL HEALTH MONITORING AND INFORMATION SYSTEMS FOR SUSTAINABLE SOIL MANAGEMENT IN THE MEDITERRANEAN REGION

WP2. Soil DATA and soil information systems (SIS) in PRIMA countries

Task 2.1. Legacy data

Sub-Task: Legacy Soil Profile Data Models for the Soil Information System

Instructions for the compilation of legacy soil profile xlsx template

Version 1.0_2024

The Soil Information System (SIS) that SOILS4MED implements needs to have the capacity to collect all the soil legacy data gathered over time by the partner countries according diverse methods and formats, and at the same time provide a sound basis for data harmonization, interoperability, and intuitive consultation. For this reason we decided to adopt the field guide proposed by WRB 4TH EDITION, which can be considered as the international state-of-the-art reference.

To maximize the advantages of the use of this method some changes. In this document we explain in detailed all the proposed changes that mostly concern the insertion of new fields, the modification of some pre-existing ones and small changes to the attributes associated with some fields.

The changes implemented are reflected in the WRB soil profile description sheet attached to this document (excel file named WRB_soil_description_sheet_for_SOILS4MED_project_2024).

1. General information

- In addition to the two sheets originally included in the excel table provided by the WRB (i.e., General and surface and Layer descriptions) two other sheets have been added: one dedicated to laboratory analyzes (Lab data) and another dedicated to classification (Soil classification).
- In the original two sheets of the Excel file, we added or modify many fields: all newly inserted fields are shown with a blue background, while the fields that have undergone modifications have the modified portions written in red.
- In the case of insertion of new fields, all the information useful for compilation is provided. If the inserted field are already described in other guides, the description and information provided by the manual taken as reference is also reported.

2. List of changes made to the field guide proposed by WRB 4TH EDITION and detailed indications for the excel WRB soil profile description compilation

All the changes made and the information useful for filling in the fields of the 4 excel tables are listed below.

Sheet 1. "General and surface"

Chapter 8.2.1

- New field: Profile code

Enter a profile identification code, different for each profile.
The code must contain both letters and numbers. In particular, the code must consist of three letters and three numbers (e.g., AND001).
IMP: the code must be different for each profile.

- New field: Method used for survey

Enter the method used to carry out the field survey in order to allow correct interpretation of the information entered.
See the table 1 to select the correct code.
Table 1. Method used for soil survey.

Method used for soil survey	Code	Year	Code
USDA soil survey manual	US	1952	US52
		1962	US62
		1975	US75
		1993	US93
		2018	US18
FAO	FA	1952	FA52
		1977	FA77
		1990	FA90
		2006	FA06
WRB	WR	2022	WR22
CPCS	CP		
OTHER	OTH		
UNKNOWN	UN		

Chapter 8.2.2

- Modified field: GPS coordinates in WGS84 SYSTEM

"in WGS 84 system" has been added to the field description.
NB: All coordinates must be entered in the WGS84 reference system and must be in decimal degree for minutes and seconds.
The following rule should be used to convert a DMS coordinate to a decimal degree:
DMS = dd°mm'ss.sss''N DECIMAL DEGREE = dd + mm/60 + ss.sss/3600
DMS = dd°mm'ss.sss''S DECIMAL DEGREE = -dd + mm/60 + ss.sss/3600
DMS = dd°mm'ss.sss''E DECIMAL DEGREE = dd + mm/60 + ss.sss/3600
DMS = dd°mm'ss.sss''W DECIMAL DEGREE = -dd + mm/60 + ss.sss/3600

- New field: GPS

Field inserted to define whether the profile position are acquired from a map or taken with the GPS tracker
See the table 2 to select the correct code.
Table 2. Information about GPS data.

GPS data	Code
Profile position taken using a gps tracker	Yes
Profile position acquired from a map	No
Unknown	Un

- Modified field: Elevation a.s.l.

"Elevation" replaced the term "Altitude" due to semantic accuracy.
The field description is unchanged

- New field: Elevation from DEM a.s.l.

Automatically filling field starting from the DEM.

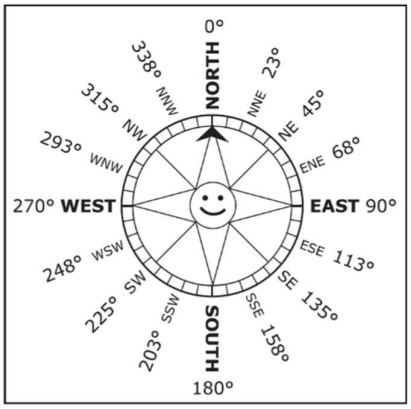
Chapter 8.2.3

- Info for the compilation: Gradient upslope & gradient downslope

  If the gradient value available is related to the entire slope, enter that in both fields.
- Info for the compilation: Slope aspect

  The aspect value must be entered in degrees, in case of data with the cardinal points translate into degrees according to the Figure 1. 8.7 of field guide WRB 4TH EDITION (Annex 1), inserted below.

Figure 1. Slope aspect, Figure 8.7 of WRB 4TH EDITION (Annex 1) (Schoeneberger et al. (2012), 1-5)



Chapter 8.2.3.bis (NEW) Geomorphic features

In this new sub-chapter the landform affecting the area of the profile and which may have influenced its formation and evolution must be indicated.
Two types of landforms ("Landform 1" and "Landform 2") and their degree of activity ("Activity of landform 1" and "Activity of landform 2") can be reported. Moreover, a general description of the landscape can be inserted in the field "Description of geomorphic features", if the data is available.
Below the instructions for filling in the fields.

- New fields: Landform 1 and Landform 2

Select among the landforms in table 3 those affecting the area of interest and enter the corresponding code. Use the help of a geomorphological map.
If the landform type is unclear, just use the general category (e.g. 'A' for Anthropogenic landforms; 'K' for Karst landforms; etc.).
IMP. If you only have a description of the morphology and do not find a match with the proposed landforms, you can only fill in the field "Description of geomorphic features"

Table 3.

Code	**Description**		

UN	**Unknown**		
A	**Anthropogenic landforms**		
AA	Area of accumulation	AAW	Landfill of anthropic origin waste
		AAR	Landfill of remoulded natural material
AM	Mining area	AMQ	Abandoned quarry
		AMA	Active quarry
AC	Embankment or other constructions for water stream		
AT	Terraced slope	ATD	Deteriorated terraced slope
		ATI	Intact terraced slope
		ATM	Mechanizable terraced slope, with connected levels
AR	Grading, reshaped slope		
AS	Anthropic sinkhole		
K	**Karst landforms**		
KD	Karst depression	KDD	Sinkhole (Doline)
		KDU	Uvala
		KDP	Polje
		KDR	Residual relief (hum)
KK	Karren (field), grooves		
KG	Griza		
KF	Fluvio-karst valley	KFB	Blind valley
		KFC	Pocket valley
		KFD	Dry valley
KC	Cave		
S	**Slope landforms**		
SD	Slope deposit (talus)		
SL	Translational/ Rotational Landslide	SLN	Landslide niche
		SLD	Landslide deposit
SR	Rockfall and toppling	SRN	Landslide niche
		SRD	Landslide deposit
SF	Debris/Mud flow deposit		
SA	Deep-seated gravitational slope deformations (trenches, counter slopes)		
SC	Soil creep terraces		
SS	Solifluction lobes		
SG	Gelifluction lobes		
SO	Soil slips deposit		
W	**Water related landforms**		
WH	Hanging valley step		
WV	V shaped small Valley		
WW	Wide fluvial valley		
WF	Flat bottom fluvial valley		
WG	Gorge		

WT	Fluvial terrace	WTE	Erosion scarp
		WTS	Surface
WL	Rills and gullies		
WB	Badlands	WBC	Calanco
		WBB	Biancane
WP	Earth pyramids		
WC	Glacis		
WA	Alluvial fan		
WD	Colluvial deposit		
WU	Alluvial plain	WUB	Reclaimed alluvial plain
		WUD	Depressed area in an alluvial plain
WR	Riverbed	WRB	Braided
		WRA	Anastomosing
		WRM	Meandering
		WRS	Single channel
		WRE	Ephemeral
		WRC	Crevasse splay
		WRR	River bank
		WRI	Fluvial island
		WRF	Floodplain and overflow area
WS	Spring depression		
G	**Glacial and periglacial landforms**		
GB	Block stream/field		
GR	Rock glacier		
GC	Glacier cirque		
GU	Crioturbation unevenness		
GH	Hummock		
GP	Poligonal soil		
GF	Fluvio-glacial deposit	GFE	Esker
		GFK	Kame terrace
		GFS	Sandur
GV	Glacial valley	GVH	Hanging glacial valley
GD	Glacial deposit	GDS	Subglacial deposit
		GDA	Ablation moraine
		GDF	Lateral/Frontal moraine
		GDI	Intermoraine depression
		GDD	Drumlin
GN	Nivations niche		
GS	Bowl shaped depression	GSI	Infilled bowl shaped depression
GT	Glacial terrace		
GA	Snow Avalanche channel	GAC	channel
		GAD	deposit
GO	Protalus ramparts		

M	**Marine and Lacustrine landforms**		
MA	Abrasion platform	MAC	Cliff foot (talus)
ML	Lacustrine terrace		
MP	Lacustrine plain	MPM	Mineral deposit infill
		MPO	Organic deposit infill (peat bog)
MR	Reclaimed draining lacustrine plain	MRM	Mineral deposit infill
		MRO	Organic deposit infill (peat bog)
MC	Coastal plain	MCT	Tidal plain
		MCR	Coastal ribbon
		MCM	Mud plain
		MCC	Tidal canal
		MCS	Sand plain
MT	Marine terrace		
MO	Lagoon		
MD	River delta	MDR	Reclaimed delta
T	**Structural landforms**		
TC	Cuesta		
TD	Structural depression (graben)		
TG	Irregular slopes due to tectonic structure influence		
TR	Structural relief (horst)		
TS	Structural surface	TSD	Dissected structural surface
		TSW	Wavy structural surface
TV	Fault slope		
V	**Volcanic landforms**		
VA	Caldera		
VR	Volcanic crater	VRM	Maar crater
VC	Volcanic cone	VCA	Volcanic ash cone
		VCL	Volcanic lava cone
		VCP	Polygenic volcanic cone
		VCS	Scoria volcanic cone
VD	Lava dome		
VL	Lava flow		
VH	Lahar deposit		
VF	Pyroclastic flow deposit		
VP	Volcanic Plateau		
VS	Structural – Volcanic depression		
E	**Aeolian landform**		
ES	Aeolian deflation surface		
ED	Aeolian deposit	EDL	Loess
		EDE	Dune affected by erosion
		EDD	Dune

		EDS	Stabilized dune
		EDA	Dune affected by anthropic influence
EI	Interdune area	EIL	Interdune area periodically inundated

- New fields: Activity of Landform 1 and Activity of Landform 2

If available, insert the code corresponding to the degree of activity of processes that shaped the landscape (table 4).

Table 4. Landform degree of activity.

Degree of activity	Code
Active or Quiescent	A
Inactive	I
Relict	R
Unknown	UN

IMP: In instances where definitive data concerning landform activity is unavailable (i.e. in the absence of geomorphological maps or field evidence), entering 'UN' or leaving the field empty.

- New field: Description of geomorphic features

Insert a brief description of the landforms and the related processes that have shaped the landscape in the past and today

Chapter 8.2.4

- New field: soil temperature regime

The field and its description are taken from the guidelines for soil description, FAO, 2006

Report the code of soil temperature regime according to table 5.

"Where applicable, the soil climate classification should be indicated.

The soil moisture and temperature regimes according to Keys to Soil Taxonomy (USDA Soil Survey Staff, 2003) may be mentioned (Table 3; explanations in Annexes 1 and 2). Where such information is not available or cannot be derived from representative climate data with confidence, it is preferable to leave the space blank. Other agroclimate parameters worth mentioning would be a local climate class, the agroclimate zone, length of growing period, etc.

Note for classification purposes

→ Soil temperature < 0 °C (pergelic soil temperature regime) → cryic horizon and Gelic qualifier." (FAO, 2006)

Annex 1 Explanation of soil temperature regimes (Adapted from USDA, 1999) (FAO, 2006)

The temperature of a soil is one of its important properties. Within limits, temperature controls the possibilities for plant growth and for soil formation. Below freezing point, there is no biotic activity, water no longer moves as a liquid, and, unless there is frost heaving, time stands still for the soil. Between 0 and 5 °C, root growth of most plant species and germination of most seeds are impossible.

A horizon as cold as 5 °C is a thermal pan to the roots of most plants.

Each pedon has a characteristic temperature regime that can be measured and described. For most practical purposes, the temperature regime can be described by: the mean annual soil temperature; the average seasonal fluctuations from that mean; and the mean warm or cold seasonal temperature gradient within the main rootzone, which is the zone from a depth of 5 to 100 cm.

MEAN ANNUAL SOIL TEMPERATURE

Each pedon has a mean annual temperature that is essentially the same in all horizons at all depths in the soil and at depths considerably below the soil. The measured mean annual soil temperature is seldom the same in successive depths at a given location. However, the differences are so small that it seems valid and useful to take a single value as the mean annual temperature of a soil.

The mean annual soil temperature is related most closely to the mean annual air temperature, but this relationship is affected to some extent by: the amount and distribution of rain; the amount of snow; the protection provided by shade

and by O horizons in forests; the slope aspect and gradient; and irrigation. Other factors, such as soil colour, texture, and content of organic matter, have negligible effects.

FLUCTUATIONS IN SOIL TEMPERATURE

The mean annual temperature of a soil is not a single reading but the average of a series of readings. Near the surface, the readings may fluctuate from the mean fully as much as those of the air temperature, especially where there is no insulating cover. The fluctuations occur as daily and annual cycles, which weather events make somewhat irregular in most places. The fluctuations decrease with increasing depth and are ultimately damped out in the substrata in a zone where the temperature is constant and is the same as the mean annual soil temperature.

ESTIMATION OF SOIL TEMPERATURE

Soil temperature can often be estimated from climatological data with a precision that is adequate for the present needs of soil surveys. Where it is not possible to make reasonably precise estimates, the measurement of soil temperature need not be a difficult or a time-consuming task.

Frequently, the mean annual soil temperature for much of the United States of America is estimated by adding 1 °C to the mean annual air temperature.

The mean summer soil temperature at a specific depth can also be estimated.

To make this estimate, it is possible to take the average summer temperatures of the upper 100 cm and correct for the temperature-depth gradient by adding or subtracting 0.6 °C for each 10 cm above or below a depth of 50 cm. The mean winter temperature of many mid-latitude soils can be estimated from the difference between the mean annual temperatures and the mean summer temperatures because the differences are of the same magnitude but have opposite signs.

CLASSES OF SOIL TEMPERATURE REGIMES

The following is a description of the soil temperature regimes used in defining classes at various categoric levels in the soil taxonomy of the United States of America.

PG – Pergelic (Latin per, throughout in time and space, and gelare, to freeze; meaning permanent frost). Soils with a pergelic temperature regime have a mean annual temperature lower than 0 °C. These are soils that have permafrost if they are moist, or dry frost if there is no excess water.

CR – Cryic (Greek kryos, coldness; meaning very cold soils). Soils in this temperature regime have a mean annual temperature lower than 8 °C but do not have permafrost.

1. In mineral soils, the mean summer soil temperature (June, July and August in the Northern Hemisphere, and December, January, and February in the Southern Hemisphere) either at a depth of 50 cm from the soil surface or at a densic, lithic or paralithic contact, whichever is shallower, is as follows:

a. if the soil is not saturated with water during some part of the summer

and:

(1) If there is no O horizon: lower than 15 °C; or

(2) If there is an O horizon: lower than 8 °C; or

b. if the soil is saturated with water during some part of the summer and:

(1) If there is no O horizon: lower than 13 °C; or

(2) If there is an O horizon or a histic epipedon: lower than 6 °C.

2. In organic soils, the mean annual soil temperature is lower than 6 °C.

Cryic soils that have an aquic moisture regime are commonly churned by frost.

Isofrigid soils could also have a cryic temperature regime. A few with organic materials in the upper part are exceptions.

The concepts of the soil temperature regimes described below are used in defining classes of soils in the low categories.

FR – Frigid

A soil with a frigid temperature regime is warmer in summer than a soil with a cryic regime. However, its mean annual temperature is lower than 8 °C and the difference between mean summer (June, July and August) and mean winter (December, January and February) soil temperatures is more than 6 °C either at a depth of 50 cm from the soil surface or at a densic, lithic or paralithic contact, whichever is shallower.

IF – Isofrigid

The mean annual soil temperature is lower than 8 °C and the mean summer and mean winter soil temperatures differ by less than 6 °C at a depth of 50 cm or at a densic, lithic or paralithic contact, whichever is shallower.

ME – Mesic

The mean annual soil temperature is 8 °C or higher but lower than 15 °C, and the difference between mean summer and mean winter soil temperatures is more than 6 °C either at a depth of 50 cm from the soil surface or at a densic, lithic or paralithic contact, whichever is shallower.
IM – Isomesic
The mean annual soil temperature is 8 °C or higher but lower than 15 °C and the mean summer and mean winter soil temperatures differ by less than 6 °C at a depth of 50 cm or at a densic, lithic or paralithic contact, whichever is shallower.
TH – Thermic
The mean annual soil temperature is 15 °C or higher but lower than 22 °C, and the difference between mean summer and mean winter soil temperatures is more than 6 °C either at a depth of 50 cm from the soil surface or at a densic, lithic or paralithic contact, whichever is shallower.
IT – Isothermic
The mean annual soil temperature is 15 °C or higher but lower than 22 °C and the mean summer and mean winter soil temperatures differ by less than 6 °C at a depth of 50 cm or at a densic, lithic or paralithic contact, whichever is shallower.
HT – Hyperthermic
The mean annual soil temperature is 22 °C or higher, and the difference between mean summer and mean winter soil temperatures is more than 6 °C at a depth of 50 cm from the soil surface or at a densic, lithic or paralithic contact, whichever is shallower.
IH – Isohyperthermic
The mean annual soil temperature is 22 °C or higher and the mean summer and mean winter soil temperatures differ by less than 6 °C at a depth of 50 cm or at a densic, lithic or paralithic contact, whichever is shallower.

Table 5. Soil temperature regime codes, table 3 of guidelines for soil description, FAO, 2006

Soil temperature regime			
PG	= Pergelic		
CR	= Cryic		
FR	= Frigid	IF	= Isofrigid
ME	= Mesic	IM	= Isomesic
TH	= Thermic	IT	= Isothermic
HT	= Hyperthermic	IH	= Isohyperthermic

- <u>New field: soil moisture regime</u>

The field and its description are taken from the guidelines for soil description, FAO, 2006
Report the code of soil temperature regime according to table 6.

"Where applicable, the soil climate classification should be indicated.
The soil moisture and temperature regimes according to Keys to Soil Taxonomy (USDA Soil Survey Staff, 2003) may be mentioned. Where such information is not available or cannot be derived from representative climate data with confidence, it is preferable to leave the space blank. Other agroclimate parameters worth mentioning would be a local climate class, the agroclimate zone, length of growing period, etc.
Note for classification purposes
→ Soil temperature < 0 °C (pergelic soil temperature regime) → cryic horizon and Gelic qualifier." (FAO, 2006)

Annex 2 Explanation of soil moisture regimes (Adapted from USDA, 1999) (FAO, 2006)
The term "soil moisture regime" refers to the presence or absence either of groundwater or of water held at a tension of less than 1 500 kPa (pF 4.2) in the soil or in specific horizons during periods of the year. Water held at a tension of 1 500 kPa or more is not available to keep most mesophytic plants alive. The availability of water is also affected by dissolved salts. Where a soil is saturated with water that is too salty to be available to most plants, it is considered salty rather than dry. Consequently, a horizon is considered dry when the moisture tension is 1 500 kPa or more, and it is considered moist when water is held at a tension of less than 1 500 kPa but more than zero. A soil may be continuously moist in some or all horizons either throughout the year or for some part of the year. It may be either moist in winter and dry in summer or the reverse. In the Northern Hemisphere, summer refers to June, July and August, and winter refers to December, January and February.

SIGNIFICANCE TO SOIL CLASSIFICATION

The moisture regime of a soil is an important property of the soil as well as a determinant of processes that can occur in the soil. During geological time, there have been significant changes in climate. Soils that could have formed only in a humid climate are now preserved in an arid climate in some areas. Such soils have relict features that reflect the former moisture regime and other features that reflect the present moisture regime.

Each of the moisture regimes in the history of a soil is a factor in the genesis of that soil and is the cause of many accessory characteristics. However, most of the accessory characteristics and those most important for interpretations are associated with the present moisture regime, even if the present regime differs widely from some of the earlier. More importantly, the present climate determines use and management of the soil. It is a property of the soil. Furthermore, the moisture regimes of most soils are inferred from the present climate, and small-scale maps can be interpreted in terms of the many accessory characteristics that are common to most of the soils that have a common climate. These characteristics include: the amount, nature and distribution of organic matter; the base status of the soil; and the presence or absence of salts.

NORMAL YEARS

In the discussions that follow and throughout the keys, the term "normal years" is used. A normal year is defined as a year that has plus or minus one standard deviation of the long-term mean annual precipitation (long term refers to 30 years or more). In addition, the mean monthly precipitation in a normal year must be plus or minus one standard deviation of the long-term monthly precipitation for 8 of the 12 months. Normal years can usually be calculated from the mean annual precipitation. However, when catastrophic events occur during a year, the standard deviations of the monthly means should also be calculated.

ESTIMATION

The landscape position of every soil is subject to extremes in climate. While no two years have exactly the same weather conditions, the moisture status of the soil must be characterized by probability. Weather probabilities can be determined from long-term weather records and observations of how each soil responds to weather conditions as modified by its landscape position. A number of methods have been devised to relate soil moisture to meteorological records. To date, all these methods have some shortcomings, even for gently sloping soils that depend primarily on precipitation for their moisture. Dew and fog can add appreciable amounts of moisture to some soils, but quantitative data are rare.

SOIL MOISTURE CONTROL SECTION

The intention in defining the soil moisture control section is to facilitate estimation of soil moisture regimes from climate data. The upper boundary of this control section is the depth to which a dry (tension of more than 1 500 kPa, but not air-dry) soil will be moistened by 2.5 cm of water within 24 hours. The lower boundary is the depth to which a dry soil will be moistened by 7.5 cm of water within 48 hours. These depths do not include the depth of moistening along any cracks or animal burrows that are open to the surface.

The boundaries for the soil moisture control section correspond to the rooting depths for many crops. However, there are natural plant communities that have their roots either above or below the control section. Attempts are currently being made to improve the parameters of the soil moisture control section.

If 7.5 cm of water moistens the soil to a densic, lithic, paralithic or petroferric contact or to a petrocalcic or petrogypsic horizon or a duripan, the contact or the upper boundary of the cemented horizon constitutes the lower boundary of the soil moisture control section.

The concept of the soil moisture control section does not apply well to the cracking clays, because these clays remoisten from both the surface and the bases of the cracks. The soil moisture patterns of these soils are defined in terms of the pattern of cracking over time. If moistening occurs unevenly, the weighted average depth of moistening in a pedon is used for the limits of the moisture control section.

The moisture control section of a soil extends approximately:

→ from 10 to 30 cm below the soil surface if the particle-size class of the soil is fine-loamy (> 15 percent particles 0.1–75 mm and 18–35 percent clay), coarse-silty (< 15 percent particles 0.1–75 mm and < 18 percent clay in fine earth), fine-silty (< 15 percent particles 0.1–75 mm and 18–35 percent clay in fine earth), or clayey (> 35 percent clay);

→ from 20 to 60 cm if the particle-size class is coarse-loamy silty (>15 percent particles 0.1–75 mm and < 18 percent clay in fine earth);

→ from 30 to 90 cm if the particle-size class is sandy (texture of sand or loamy sand).

If the soil contains rock and pararock fragments that do not absorb and release water, the limits of the moisture control section are deeper. The limits of the soil moisture control section are affected not only by the particle-size class but also by differences in soil structure or pore-size distribution or by other factors that influence the movement and retention of water in the soil.

CLASSES OF SOIL MOISTURE REGIMES

The soil moisture regimes are defined in terms of the level of groundwater and in terms of the seasonal presence or absence of water held at a tension of less than 1 500 kPa in the moisture control section. It is assumed in the definitions that the soil supports whatever vegetation it is capable of supporting, i.e., crops, grass or native vegetation, and that

the amount of stored moisture is not being increased by irrigation or fallowing. These cultural practices affect the soil moisture conditions as long as they continued.

AQ – Aquic moisture regime

The aquic (Latin aqua, water) moisture regime is a reducing regime in a soil that is virtually free of dissolved oxygen because it is saturated by water. Some soils are saturated with water at times while dissolved oxygen is present, either because the water is moving or because the environment is unfavourable for micro-organisms

(e.g. where the temperature is less than 1 °C); such a regime is not considered aquic.

It is not known how long a soil must be saturated before it is said to have an aquic moisture regime. However, the duration must be at least a few days, because it is implicit in the concept that dissolved oxygen is virtually absent.

PQ – Peraquic moisture regime

Very commonly, the level of groundwater fluctuates with the seasons. However, there are soils in which the groundwater is always at or very close to the surface. Examples are soils in tidal marshes or in closed, landlocked depressions fed by perennial streams. Such soils are considered to have a peraquic moisture regime.

AR – Aridic and TO – torric (Latin aridus, dry, and torridus, hot and dry) moisture regimes

These terms are used for the same moisture regime but in different categories of the taxonomy.

In the aridic (torric) moisture regime, the moisture control section in normal

years is both:

→ dry in all parts for more than half on the cumulative days per year when the soil temperature at a depth of 50 cm from the soil surface is above 5 °C;

→moist in some or all parts for less than 90 consecutive days when the soil temperature at a depth of 50 cm is above 8 °C.

Soils that have an aridic (torric) moisture regime normally occur in areas of arid climates. A few are in areas of semi-arid climates and either have physical properties that keep them dry, such as a crusty surface that virtually precludes the infiltration of water, or are on steep slopes where runoff is high. There is little or no leaching in

this moisture re

In tropical and subtropical regions that have a monsoon climate with either one or two dry seasons, summer and winter seasons have little meaning. In such regions, the moisture regime is ustic if there is at least one rainy season of three months or more. In temperate regions of subhumid or semi-arid climates, the rainy seasons are usually spring and summer or spring and autumn, but never winter. Native plants are mainly annuals or plants that have a dormant period while the soil is dry.

XE – Xeric moisture regime

The xeric (Greek xeros, dry) moisture regime is the typical moisture regime in areas of Mediterranean climates, where winters are moist and cool and summers are warm and dry. The moisture, which falls during the winter, when potential evapotranspiration is at a minimum, is particularly effective for leaching.

In areas of a xeric moisture regime, the soil moisture control section, in normal years, is dry in all parts for 45 or more consecutive days in the 4 months following the summer solstice and moist in all parts for 45 or more consecutive days in the 4 months following the winter solstice. Moreover, in normal years, the moisture control section is moist in some part for more than half of the cumulative days per year when the soil temperature at a depth of 50 cm from the soil surface is higher than 6 °C or for 90 or more consecutive days when the soil temperature at a depth of 50 cm is higher than 8 °C. The mean annual soil temperature is lower than 22 °C, and the mean summer and mean winter soil temperatures differ by 6 °C or more either at a depth of 50 cm from the soil surface or at a densic, lithic or paralithic contact if shallower.

Table 6. Soil moisture regime codes, table 3 of guidelines for soil description, FAO, 2006

Soil moisture regime			
AQ	= Aquic	PQ	= Peraquic
DU	= Udic	PU	= Perudic
US	= Ustic		
XE	= Xeric		
[illegible]	= Aridic/ariu[illegible]	[illegible]	= Torric

Chapter 8.2.5

- New field: Land Use

The field and its description are taken from the guidelines for soil description, FAO, 2006 (without additional codes).

Report the Land Use classification according to table 7.

"Land use applies to the current use of the land, whether agricultural or nonagricultural, in which the soil is located. Land use has a major influence on the direction and rate of soil formation; its recording enhances the interpretative value of the soil data considerably." (FAO, 2006)

Table 7. Land use classification, table 8 of guidelines for soil description, FAO, 2006

Land-use classification

A	**= Crop agriculture (cropping)**		
	AA	= Annual field cropping	
		AA1	= Shifting cultivation
		AA2	= Fallow system cultivation
		AA3	= Ley system cultivation
		AA4	= Rainfed arable cultivation
		AA5	= Wet rice cultivation
		AA6	= Irrigated cultivation
	AP	= Perennial field cropping	
		AP1	= Non-irrigated cultivation
		AP2	= Irrigated cultivation
	AT	= Tree and shrub cropping	
		AT1	= Non-irrigated tree crop cultivation
		AT2	= Irrigated tree crop cultivation
		AT3	= Non-irrigated shrub crop cultivation
		AT4	= Irrigated shrub crop cultivation
M	**= Mixed farming**		
	MF	= Agroforestry	
	MP	= Agropastoralism	
H	**= Animal husbandry**		
	HE	= Extensive grazing	
		HE1	= Nomadism
		HE2	= Semi-nomadism
		HE3	= Ranching
	HI	= Intensive grazing	
		HI1	= Animal production
		HI2	= Dairying
F	**= Forestry**		
	FN	= Natural forest and woodland	
		FN1	= Selective felling
		FN2	= Clear felling
	FP	= Plantation forestry	
P	**= Nature protection**		
	PN	= Nature and game preservation	
		PN1	= Reserves
		PN2	= Parks
		PN3	= Wildlife management
	PD	= Degradation control	
		PD1	= Without interference
		PD2	= With interference
S	**= Settlement, industry**		
	SR	= Residential use	
	SI	= Industrial use	
	ST	= Transport	
	SC	= Recreational use	
	SX	= Excavations	
	SD	= Disposal sites	
Y	**= Military area**		
O	**= Other land uses**		
U	**= Not used and not managed**		

- <u>New field: Corine Land Cover Class</u>

Report the Corine Land Cover Class according to last version available.

- <u>New field: "Vegetation cover (%)" in Cultivated land</u>

Report the vegetation cover. For the upper stratum and the mid-stratum, report the percentage (by area) of the crown cover. For the ground stratum, report the percentage (by area) of the ground cover.

<u>Chapter 8.3.3 bis</u>

In this new sub-chapter the bedrock of the area should be described, reporting the lithology and the formation name, if available.

- <u>New field: Geological Formation name</u>

Report the name of the geological formation

- New field: Lithology

Report the lithology type according to the table X.

Table X. Lithology type. Table modified from table 8.91 WRB 4TH EDITION (Annex 1).

Major class	Code	Group	Code	Type	Code
Igneous Rock	I	Felsic igneous	IF	Granite	IF1
				Quartz-diorite	IF2
				Grano-diorite	IF3
				Diorite	IF4
				Rhyolite	IF5
		Intermediate igneous	II	Andesite, trachyte, phonolite	II1
				Diorite-syenite	II2
		Mafic igneous	IM	Gabbro	IM1
				Basalt	IM2
				Dolerite	IM3
		Ultramafic igneous	IU	Peridotite	IU1
				Pyroxenite	IU2
				Serpentinite	IU3
		Pyroclastic	IP	Tuff, tuffite	IP1
				Volcanic scoria/breccia	IP2
				Volcanic ash	IP3
				Ignimbrite	IP4
Metamorphic rock	M	Felsic metamorphic	MF	Quartzite	MF1
				Gneiss, migmatite	MF2
				Slate, phyllite (pelitic rocks)	MF3
				Schist	MF4
		Mafic metamorphic	MM	Slate, phyllite (pelitic rocks)	MM1
				(Green)schist	MM2
				Gneiss rich in Fe-Mg minerals	MM3
				Metamorphic limestone (marble)	MM4
				Amphibolite	MM5
				Eclogite	MM6
		Ultramafic metamorphic	MU	Serpentinite, greenstone	MU1
Sedimentary rock (consolidated)	S	Clastic sediments	SC	Conglomerate, breccia	SC1
				Sandstone, greywacke, arkose	SC2
				Silt-, mud-, claystone	SC3
				Shale	SC4
				Ironstone	SC5
		Carbonatic, organic	SO	Limestone, other carbonate rock	SO1
				Marl and other mixtures	SO2
				Coals, bitumen and related rocks	SO3
		Evaporites	SE	Anhydrite, gypsum	SE1
Sedimentary rock (unconsolidated)	U	Weathered residuum	UR	Bauxite, laterite	UR1
		Fluvial	UF	Sand and gravel	UF1
				Clay, silt and loam	UF2
		Lacustrine	UL	Sand	UL1

				Silt and clay, < 20% $CaCO_3$ equivalent, little or no diatoms	UL2
				Silt and clay, < 20% $CaCO_3$ equivalent, many diatoms	UL3
				Silt and clay, ≥ 20% $CaCO_3$ equivalent (marl)	UL4
		Marine, estuarine	UM	Sand	UM1
				Clay and silt	UM2
		Colluvial	UC	Slope deposits	UC1
				Lahar	UC2
				Deposit of soil material	UC3
		Aeolian	UE	Loess	UE1
				Sand	UE2
		Glacial	UG	Moraine	UG1
				Glacio-fluvial sand	UG2
				Glacio-fluvial gravel	UG3
		Cryogenic	UK	Periglacial rock debris	UK1
				Periglacial solifluction layer	UK2
		Organic	UO	Rainwater-fed peat (bog)	UO1
				Groundwater-fed peat (fen)	UO2
				Lacustrine (organic limnic sediments)	UO3
		Anthropogenic/ technogenic	UA	Redeposited natural material	UA1
				Industrial/artisanal deposits	UA2
		Unspecified deposits	UU	Clay	UU1
				Loam and silt	UU2
				Sand	UU3
				Gravelly sand	UU4
				Gravel, broken rock	UU5

Chapter 8.3.4

- Modified field: "Size class" in Coarse surface fragments

The table 8.11 in the field guide proposed by WRB 4TH EDITION (Annex 1) has been modified inserting a new value of size class: "Gravel". The code "Gravel" may be used if the data collected do not distinguish between fine, medium and coarse gravel.

Report the size of coarse fragments according to table X.

Table X. Size of coarse surface fragments, Modified version of Table 8.11 of the field guide WRB 4TH EDITION (Annex 1), FAO (2006), Table 15

Size (cm)	Size class	Code	Size class	Code
> 0.2 - 0.6	Fine gravel	F	Gravel	G
> 0.6 - 2	Medium gravel	M		
> 2 - 6	Coarse gravel	C		
> 6 - 20	Stones	S		
> 20 - 60	Boulders	B		
> 60	Large boulders	L		
No coarse surface fragments	N			

Chapter 8.3.9

- New field: Drainage conditions

The field and its description are taken from the Soil Survey Manual, USDA, 2018.

Report the code of the soil drainage conditions class (Table 8).

"Natural drainage class refers to the frequency and duration of wet periods under conditions similar to those under which the soil developed. Alteration of the water regime by humans, either through drainage or irrigation, is not a consideration unless the alterations have significantly changed the morphology of the soil. Descriptions of these classes follow.

Excessively drained.—Water is removed very rapidly. Internal free water occurrence commonly is very rare or very deep. The soilsare commonly coarse textured and have very high saturated hydraulic conductivity or are very shallow.

Somewhat excessively drained.—Water is removed from the soil rapidly. Internal free water occurrence commonly is very rare or very deep. The soils are commonly coarse textured and have high saturated hydraulic conductivity or are very shallow.

Well drained.—Water is removed from the soil readily but not rapidly. Internal free water occurrence commonly is deep or very deep; annual duration is not specified. Water is available to plants throughout most of the growing season in humid regions. Wetness does not inhibit root growth for significant periods during most growing seasons. The soils are mainly free of, or are deep or very deep to, redoximorphic features related to wetness.

Moderately well drained.—Water is removed from the soil somewhat slowly during some periods of the year. Internal free water occurrence is commonly moderately deep and transitory through permanent. The soils are wet for only a short time within the rooting depth during the growing season but long enough that most mesophytic crops are affected. They commonly have a moderately low or lower saturated hydraulic conductivity in a layer within the upper 1 meter, periodically receive high rainfall, or both.

Somewhat poorly drained.—Water is removed slowly so that the soil is wet at a shallow depth for significant periods during the growing season. Internal free water occurrence is commonly shallow to moderately deep and transitory to permanent. Wetness markedly restricts the growth of mesophytic crops, unless artificial drainage is provided. The soils commonly have one or more of the following characteristics: low or very low saturated hydraulic conductivity, a high water table, additional water from seepage, or nearly continuous rainfall.

Poorly drained.—Water is removed so slowly that the soil is wet at shallow depths periodically during the growing season or remains wet for long periods. Internal free water occurrence is shallow or very shallow and common or persistent. Free water is commonly at or near the surface long enough during the growing season that most mesophytic crops cannot be grown, unless the soil is artificially drained. The soil, however, is not continuously wet directly below plow depth. Free water at shallow depth is common. The water table is commonly the result of low or very low saturated hydraulic conductivity, nearly continuous rainfall, or a combination of these.

Very poorly drained.—Water is removed from the soil so slowly that free water remains at or very near the surface during much of the growing season. Internal free water occurrence is very shallow and persistentor permanent. Unless the soil is artificially drained, most mesophytic crops cannot be grown. The soils are commonly level or depressed and frequently ponded. In areas where rainfall is high or nearly continuous, slope gradients may be greater.

Subaqueous.—Free water is above the soil surface. Internal free water occurrence is permanent, and there is a positive water potential at the soil surface for more than 21 hours of each day. The soils have a peraquic soil moisture regime."

Table 8. Drainage conditions class

Drainage conditions	code
excessively drained	ED
somewhat excessively drained	SED
well drained	WD
moderately well drained	MWD
Somewhat poorly drained	SPD
poorly drained	PD
very poorly drained	VPD
Subaqueous	SA

- New field: Depth of ground water (cm)

Report the depth of the ground water in cm

Chapter 8.3.11

- New field: “total area affected by erosion and deposition” in Surface uneveness

The field and its description are taken from the guidelines for soil description, FAO, 2006
Report the total area affected by erosion and deposition in %.

→ New field: Notes

Report in this field the important information that cannot be inserted in any previous field but could be useful.

Sheet 2. "Layer description"

Chapter 8.4.1

- New field: Profile code

Enter a profile identification code, different for each profile.
The code must contain both letters and numbers. In particular, the code must consist of three letters and three numbers (e.g., AND001).
→Use the same code inserted in the field "Profile code" of sheet 1.
IMP: the code must be different for each profile.

- Moved field: Horizon designation

For convenience the field has been moved to the beginning of the table, while in the official version it is at the end.

- Field: Lower boundary

Enter only the numeric value and no symbol (e.g. '+'). If the lower limit has not been reached, leave the field blank and enter 'U = unknown ' in the column "Distinctness of the layer's lower boundary".

- New Field: lower boundary range

As explained in FAO, 2006 "Most horizons do not have a constant depth. The variation or irregularity of the surface of the boundary is described by the topography in terms of smooth, wavy, irregular and broken. If required, ranges in depth should be given in addition to the average depth, for example 28 (25–31) cm to 45 (39–51) cm". In this case insert in this field the range of depth (e.g. 25-31).

- New field: Layer thickness

Report the thickness value of the different layers.
If not filled in the value of thickness will be inserted automatically based on the previous columns.

Chapter 8.4.7

- Modified field: size and shape

  The table 8.33 in the field guide proposed by WRB 4TH EDITION (Annex 1) has been modified inserting two new values of shape for each size class: "Subrounded" and "Flat" and inserting a code only for size classes according to the Table 27 and 28 FAO,2006.
  The general category "Gravel" is also inserted. Use this category if it is not possible to divide the gravel into size sub-groups.
  Report the size and shape of coarse material according to table 9.
  Table 9. Size and shape classes of coarse fragments Modified version of Table 8.33 of the field guide WRB 4TH EDITION (Annex 1), Tables 27 and 28, FAO (2006),

Size (cm)	Size class	Code	Shape	Code
> 0.2 - 6	Gravel	G	Rounded	GR
			Angular	GA
			Rounded and angular	GB
			Subrounded	GS
			Flat	GF
> 0.2 - 0.6	Fine gravel	F	Rounded	FR
			Angular	FA
			Rounded and angular	FB
			Subrounded	FS
			Flat	FF
> 0.6 - 2	Medium gravel	M	Rounded	MR
			Angular	MA
			Rounded and angular	MB
			Subrounded	MS
			Flat	MF

> 2 - 6	Coarse gravel	C	Rounded	CR
			Angular	CA
			Rounded and angular	CB
			Subrounded	CS
			Flat	CF
> 6 - 20	Stones	S	Rounded	SR
			Angular	SA
			Rounded and angular	SB
			Subrounded	SS
			Flat	SF
> 20 - 60	Boulders	B	Rounded	BR
			Angular	BA
			Rounded and angular	BB
			Subrounded	BS
			Flat	BF
> 60	Large boulders	L	Rounded	LR
			Angular	LA
			Rounded and angular	LB
			Subrounded	LS
			Flat	LF
None			NO	

- New field: Lithology of coarse fragments of size and shape class 1, Lithology of coarse fragments of size and shape class 2, Lithology of coarse fragments of size and shape class 3, Lithology of coarse fragments of size and shape class 4

Report the lithology of the coarse fragments according to the table 8.

Table 8. Lithology of coarse fragments, modified version of table 8.91 of the field guide WRB 4TH EDITION (Annex 1) (Modified FAO, 2006). Enter a maximum of three lithology types per cell separated by commas (e.g. IM,SE,MF)

Major class	**Code**	**Group**	**Code**	**Type**	**Code**
Igneous Rock	I	Felsic igneous	IF	Granite	IF1
				Quartz-diorite	IF2
				Grano-diorite	IF3
				Diorite	

				Schist	MF4
		Mafic metamorphic	MM	Slate, phyllite (pelitic rocks)	MM1
				(Green)schist	MM2
				Gneiss rich in Fe-Mg minerals	MM3
				Metamorphic limestone (marble)	MM4
				Amphibolite	MM5
				Eclogite	MM6
		Ultramafic metamorphic	MU	Serpentinite, greenstone	MU1
Sedimentary rock (consolidated)	S	Clastic sediments	SC	Conglomerate, breccia	SC1
				Sandstone, greywacke, arkose	SC2
				Silt-, mud-, claystone	SC3
				Shale	SC4
				Ironstone	SC5
		Carbonatic, organic	SO	Limestone, other carbonate rock	SO1
				Marl and other mixtures	SO2
				Coals, bitumen and related rocks	SO3
		Evaporites	SE	Anhydrite, gypsum	SE1

- <u>Modified field: Cementation agent in "Remnants of broken-up cemented layers"</u>

The table 8.35 in the field guide proposed by WRB 4TH EDITION (Annex 1) has been modified inserting a general code "F" for Fe oxides cementing agent. Report the cementation agent according to table X1.
Table X1. Remnants of broken-up cemented layers: cementing agent, Modified version of Table 8.35 of the field guide WRB 4TH EDITION (Annex 1).

Cementing agent	**Code**	**Code**
Secondary carbonates		CA
Secondary gypsum		GY
Secondary silica		SI
Fe oxides, predominantly inside (former) soil aggregates, no significant concentration of organic matter	F	FI
Fe oxides, predominantly on the surfaces of (former) soil aggregates, no significant concentration of organic matter		FO
Fe oxides, no relationship to (former) soil aggregates, no significant concentration of organic matter		FN
Fe oxides in the presence of a significant concentration of organic matter		FH

<u>Chapter 8.4.10</u>

- <u>Modified field: Type in "Structure"</u>

The table 8.42 in the field guide proposed by WRB 4TH EDITION (Annex 1) has been modified inserting two new code of structure type: "Single grain" and "Massive". The code "Single grain" and "Massive" may be used if the data collected do not distinguish among the different type of the above mentioned structure. Report the structure according to table Y.
Table Y. Types of structure, formation and codes, Modified version of Table 8.42 of the field guide WRB 4TH EDITION (Annex 1).

Type	**Formation**	**Code**	**Formation**	**Code**
Granular	Soil aggregate structure, natural	GR		
Subangular blocky	Soil aggregate structure, natural	BS		

Angular blocky	Soil aggregate structure, natural	BA		
Lenticular	Soil aggregate structure, natural	LC		
Wedge-shaped	Soil aggregate structure, natural	WE		
Prismatic	Soil aggregate structure, natural	PR		
Columnar	Soil aggregate structure, natural	CO		
Polyhedral	Soil aggregate structure, natural	PH		
Flat-edged	Soil aggregate structure, natural	FE		
Pseudosand/Pseudosilt	Soil aggregate structure, natural	PS		
Platy	Soil aggregate structure, natural or resulting from artificial pressure	PL		
Single grain	No structural units	S	No structural units, rock structure, inherited from the parent material	SR
			No structural units, soil structure, resulting from soil-forming processes, like loss of organic matter and/or oxides and/or clay minerals or loss of stratification	SS
Massive	No structural units	M	No structural units, rock structure, inherited from the parent material, structure not changing with soil moisture, not or only slightly chemically weathered	MR
			No structural units, rock structure, inherited from the parent material, structure not changing with soil moisture, strongly chemically weathered (e.g. saprolite)	MW
			No structural units, soil structure, present when moist and changing into soil aggregate structure when dry	MS
Stratified	No structural units, rock structure, visible stratification from sedimentation	ST		
Cloddy	Artificial structural elements	CL		

Chapter 8.4.12

- Modified field: Type in "Non matrix pores"

A new Type "Other" and related criterion have been inserted in the table, in order to allow the description of those pores not included in the types listed above.

Table 10. Types of non-matrix pores, modified version of Table 8.46 of the field guide WRB 4TH EDITION (Annex 1), Schoeneberger et al. (2012), 2-73, modified

Criterion	**Type**	**Code**
Cylindrical and elongated voids; e.g., worm tunnels	Tubular	TU
Cylindrical, elongated, branching voids; e.g., empty root channels	Dendritic Tubular	DT
Ovoid to spherical voids; e.g., solidified pseudomorphs of entrapped gas bubbles concentrated below a crust; most common in arid and semiarid environments and in permafrost soils	Vesicular	VE
Non-connected cavities, chambers; e.g., vughs; various shapes	Irregular	IG
Pores not attributable to any of the previous classes	Other	OT
No non-matrix pores		NO

Chapter 8.4.20

- Modified field: Location

A new Location "Mottles" has been inserted in Tabella 8.55, Location to select in case of mottles in the layer.

Table 11. Location of oximorphic features. Modified version of Table 8.55 of the field guide WRB 4TH EDITION (Annex 1).

Location	Code		Code
Inner parts	OI	Inside soil aggregates: masses	OIM
		Inside soil aggregates: concretions	OIC
		Inside soil aggregates: nodules	OIN
		Inside soil aggregates: both concretions and/or nodules (not possible to distinguish)	OIB
Outer parts	OO	On surfaces of soil aggregates	OOA
		Adjacent to surfaces of soil aggregates, infused into the matrix (hypocoats)	OOH
		On biopore walls, lining the entire wall surface	OOE
		On biopore walls, not lining the entire wall surface	OON
		Adjacent to biopores, infused into the matrix (hypocoats)	OOI
Random (not associated with aggregate surfaces or pores)	OR	Distributed over the layer, no order visible	ORN
		Distributed over the layer, surrounding areas with reductimorphic features	ORS
		Throughout	ORT
Mottles			OM

- New field: Contrast of mottles

Report the contrast of mottles if mottles have been selected in the location field according to the table 12.
The field and its description are taken from the guidelines for soil description, FAO, 2006
"Contrast of mottles
The colour contrast between mottles and soil matrix can be described." FAO, 2006

Table 12. Classification of the contrast of mottles, table 34 of guidelines for soil description, FAO, 2006

Classification of the contrast of mottles		
F	Faint	The mottles are evident only on close examination. Soil colours in both the matrix and mottles have closely related hues, chromas and values.
D	Distinct	Although not striking, the mottles are readily seen. The hue, chroma and value of the matrix are easily distinguished from those of the mottles. They may vary by as much as 2.5 units of hue or several units in chroma or value.
P	Prominent	The mottles are conspicuous and mottling is one of the outstanding features of the horizon. Hue, chroma and value alone or in combination are at

- New field: Boundary between mottles and matrix

Report the characteristics of boundary between mottels and matrix if mottles have been selected in the location field, according to the table 13.
The field and its description are taken from the guidelines for soil description, FAO, 2006
"Boundary of mottles
The boundary between mottle and matrix is described as the thickness of the zone within which the colour transition can be located without being in either the mottle or matrix." FAO, 2006

Table 13. Classification of boundary between mottle and matrix, table 35 of guidelines for soil description, FAO, 2006

Classification of boundary between mottle and matrix

		mm
S	Sharp	< 0.5
C	Clear	0.5–2
D	Diffuse	> 2

Chapter 8.4.23

- New field: Form of coatings

Report the form for each categories of coatings according to the table 14.
The field and its description are taken from the guidelines for soil description, FAO, 2006
"Form
For some coatings, the form may be informative for their genesis. For example, manganese and iron–manganese coatings of dendroidal form indicate their formation owing to poor infiltration and periodically reductive conditions because of percolating water."FAO,2006
Table 14. Classification of the form of coatings, table 67 of guidelines for soil description, FAO, 2006

Classification of the form of coatings

C	Continuous
CI	Continuous irregular (non-uniform, heterogeneous)
DI	Discontinuous irregular
DE	Dendroidal
DC	Discontinuous circular
O	Other

- New fields: Size and Shape

Report the size and shape for each type of secondary carbonates (i.e., "Size of Type 1", "Shape of Type 1" , "Size of Type 2", "Shape of Type 2", "Size of Type 3", "Shape of Type 3", "Size of Type 4", "Shape of Type 4") according to the table 15.
The field and its description are taken from the guidelines for soil description, FAO, 2006
Table 15. Classification of the size and shape of mineral concentrations, table 75 of guidelines for soil description, FAO, 2006

Classification of the size and shape of mineral concentrations

	Size	(mm)		Shape
V	Very fine	< 2	R	Rounded (spherical)
F	Fine	2–6	E	Elongated
M	Medium	6–20	F	Flat
C	Coarse	> 20	I	Irregular
			A	Angular

Chapet 8.4.26

- New field: Type (Type 1 and Type 2)

Report the type of secondary gypsum according to the table 16.

Table 16. Type of form of secondary gypsum.

Type of form of secondary Gypsum	**Code**
Gypsum crystal intergrowths or nodules	CR
Soft concrections	SC
Disperse powdery gypsum	DP
Filaments (vermiform gypsum, pseudomycelia)	FL

Other	OT

- New field: Size of Type 1, "Shape of Type 1, Size of Type 2 and Shape of Type 2

Report the size and shape of each type according to the table 15.

The field and its description are taken from the guidelines for soil description, FAO, 2006

Table 15. Classification of the size and shape of mineral concentrations, table 75 of guidelines for soil description, FAO, 2006

Classification of the size and shape of mineral concentrations

	Size	(mm)		Shape
V	Very fine	< 2	R	Rounded (spherical)
F	Fine	2–6	E	Elongated
M	Medium	6–20	F	Flat
C	Coarse	> 20	I	Irregular
			A	Angular

Chapter 8.4.30

- New field: Stickiness

Report the stickiness of each horizon, according to the table 17.

The field and its description are taken from the guidelines for soil description, FAO, 2006

"Stickiness is the quality of adhesion of the soil material to other objects determined by noting the adherence of soil material when it is pressed between thumb and finger." FAO, 2006

Table 17. Classification of soil stickiness, table 55 of guidelines for soil description, FAO, 2006

Classification of soil stickiness

NST	Non-sticky	After release of pressure, practically no soil material adheres to thumb and finger.
SST	Slightly sticky	After pressure, soil material adheres to both thumb and finger but comes off one or the other rather cleanly. It is not appreciably stretched when the digits are separated.
[illegible]	[illegible] sticky	After pressure, soil material adheres to both thumb and finger and tends to stretch somewhat and pull apart rather than pulling free from either digit.
'ST	Very sticky	After pressure, soil material adheres strongly to both thumb and finger and is decidedly stretched when they are separated.

Note: Additional codes are: SSS, slightly sticky to sticky; and SVS, sticky to very sticky.

Chapter 8.4.31

- New field: average thickness (cm)

Report the average thickness of the surface crusts in cm.

Chapter 8.4.37

- New fields: "Abundance of very fine roots (<0.5 mm)" and "Abundance of fine roots (0.5-2 mm)"; "Abundance of medium roots (2-5 mm)" and "Abundance of coarse roots (>5 mm)".

The diameter roots subdivision is made according to the table 18. 79, FAO, 2006.

Table 18. Classification of the diameter of roots, table 78 of guidelines for soil description, FAO, 2006

Report the abundance for each groups of roots according to Table 8.86: Abundance of roots, WRB 4TH EDITION (Annex 1)

NB: The Table 8.86 WRB 4TH EDITION (Annex 1) is not the same of that proposed by FAO (2006, Table 80)

Chapter 8.4.40

- Modified field: Parent Material

Report the parent material according table X. Use the help of a geological map.

Table X. Lithology type. Table modified from table 8.91 WRB 4TH EDITION (Annex 1).

Major class	**Code**	**Group**	**Code**	**Type**	**Code**
Igneous Rock	I	Felsic igneous	IF	Granite	IF1
				Quartz-diorite	IF2
				Grano-diorite	IF3
				Diorite	IF4
				Rhyolite	IF5
		Intermediate igneous	II	Andesite, trachyte, phonolite	II1
				Diorite-syenite	II2
		Mafic igneous	IM	Gabbro	IM1
				Basalt	IM2
				Dolerite	IM3
		Ultramafic igneous	IU	Peridotite	IU1
				Pyroxenite	IU2
				Serpentinite	IU3
		Pyroclastic	IP	Tuff, tuffite	IP1
				Volcanic scoria/breccia	IP2
				Volcanic ash	IP3
				Ignimbrite	IP4
Metamorphic rock	M	Felsic metamorphic	MF	Quartzite	MF1
				Gneiss, migmatite	MF2
				Slate, phyllite (pelitic rocks)	MF3
				Schist	MF4
		Mafic metamorphic	MM	Slate, phyllite (pelitic rocks)	MM1
				(Green)schist	MM2
				Gneiss rich in Fe-Mg minerals	MM3
				Metamorphic limestone (marble)	MM4
				Amphibolite	MM5
				Eclogite	MM6
		Ultramafic metamorphic	MU	Serpentinite, greenstone	MU1

Sedimentary rock (consolidated)	S	Clastic sediments	SC	Conglomerate, breccia	SC1
				Sandstone, greywacke, arkose	SC2
				Silt-, mud-, claystone	SC3
				Shale	SC4
				Ironstone	SC5
		Carbonatic, organic	SO	Limestone, other carbonate rock	SO1
				Marl and other mixtures	SO2
				Coals, bitumen and related rocks	SO3
		Evaporites	SE	Anhydrite, gypsum	SE1
Sedimentary rock (unconsolidated)	U	Weathered residuum	UR	Bauxite, laterite	UR1
		Fluvial	UF	Sand and gravel	UF1
				Clay, silt and loam	UF2
		Lacustrine	UL	Sand	UL1
				Silt and clay, < 20% $CaCO_3$ equivalent, little or no diatoms	UL2
				Silt and clay, < 20% $CaCO_3$ equivalent, many diatoms	UL3
				Silt and clay, ≥ 20% $CaCO_3$ equivalent (marl)	UL4
		Marine, estuarine	UM	Sand	UM1
				Clay and silt	UM2
		Colluvial	UC	Slope deposits	UC1
				Lahar	UC2
				Deposit of soil material	UC3
		Aeolian	UE	Loess	UE1
				Sand	UE2
		Glacial	UG	Moraine	UG1
				Glacio-fluvial sand	UG2
				Glacio-fluvial gravel	UG3
		Cryogenic	UK	Periglacial rock debris	UK1
				Periglacial solifluction layer	UK2
		Organic	UO	Rainwater-fed peat (bog)	UO1
				Groundwater-fed peat (fen)	UO2
				Lacustrine (organic limnic sediments)	UO3
		Anthropogenic/ technogenic	UA	Redeposited natural material	UA1
				Industrial/artisanal deposits	UA2
		Unspecified deposits	UU	Clay	UU1
				Loam and silt	UU2
				Sand	UU3
				Gravelly sand	UU4
				Gravel, broken rock	UU5

→ New field: Notes

Report in this field the important information that cannot be inserted in any previous field but could be useful.

Sheet 3. “Lab data”

General information

This sheet has been inserted ex novo to collect all the laboratory data of the profiles in the database.
Apart from the profile identification fields, the fields are divided into two types:

- Fields in which entering the numerical value of the parameter
- Fields in which entering the analytical method used to obtain the data

The method used must be selected from those proposed in the table associated with the field.
If the method used is unknown, enter "UN", i.e. unknown, and if the method used is not one of the options in the table, enter "OTH", i.e. other.

Recommendation

When entering data, pay close attention to the units of measurement used for each field.
In the case of data expressed in different units, a conversion must be carried out.

List of new fields inserted

- Profile code

Enter a profile identification code, different for each profile.
The code must contain both letters and numbers. In particular, the code must consist of three letters and three numbers (e.g., AND001).
→ Use the same code inserted in the field “Profile code” of sheet 1 and 2.

- Horizon designation

Field described in “Layer description”
→ Use the same designation inserted in the field “Horizon designation” of sheet 2.

- Number of the layer

Field described in “Layer description”
→ Use the same number inserted in the field “Number of the layer” of sheet 2.

- Gravel content

Enter the content of gravel in %

- Classification system for texture of fine earth

Report the type of classification system used according to table 19.
Table 19.

Type of classification used	Code
USDA	U
WRB	W
OTHER	OT
UNKNOWN	UN

- Sand

Enter the content of sand in % on the fine earth

- Very coarse sand

Enter the content of very coarse sand in % on the fine earth

- Coarse sand

Enter the content of coarse sand in % on the fine earth

- Medium sand

Enter the content of medium sand in % on the fine earth

- Fine sand

Enter the content of fine sand in % on the fine earth

- Very fine sand

Enter the content of very fine sand in % on the fine earth

- Method used for sand content

Report the methodology used according to table 20.

Table 20.

Method used for sand content	Code
Sieving	SV
hydrometer method	BH
Pipette and gravimetric method	PR
Laser diffraction	LD
Other	OTH
Unknown	UN

- Silt

Enter the content of silt in % on the fine earth

- Coarse silt

Enter the content of coarse silt in % on the fine earth

- Fine silt

Enter the content of fine silt in % on the fine earth

- Method used for silt content

Report the type of methodology used according to table 21.

Table 21.

Method used for silt content	Code
hydrometer method	BH
Pipette and gravimetric method	PR
Laser diffraction	LD
Other	OTH
Unknown	UN

- Clay

Enter the content of clay in % on the fine earth

- Method used for clay content

Report the type of methodology used according to table 22.

Table 22.

Method used for clay content	Code
hydrometer method	BH
Pipette and gravimetric method	PR
Laser diffraction	LD
Other	OTH
Unknown	UN

- Texture class

Report the type of texture class type according to table 23.

Table 23. Table 8.39 of the field guide WRB 4TH EDITION (Annex 1), Soil Science Division Staff (2017)

Texture class	Code
Sand (S)	S
Loamy sand	LS
Silt	Si

Silt loam	SiL
Sandy loam	SL
Loam	L
Sandy clay loam	SCL
Silty clay loam	SiCL
Clay loam	CL
Sandy clay	SC
Silty clay	SiC
Clay	C

- Bulk density

Enter the data in g/cm^3

- Electric conductivity

Enter the data in dS/m

- Method used for electric conductivity

Report the type of methodology used according to table 24.

Table 24.

Method used for electric conductivity	Code
Saturated paste	SP
Soil to water 1:2.5	OP
Soil to water 1:5	OF
Soil to water 1:1	OO
Soil to water 1:2	OT
Other	OTH
Unknown	UN

- Hydraulic conductivity at saturation

Enter the data in mm/h

- Method used for Hydraulic conductivity at saturation

Report the type of methodology used according to table 25.

Table 25.

Method used for electric conductivity	Code
Infiltrometer	IN
Other	OTH
Unknown	UN

- Saturation

Enter the data in %

- Field capacity

Enter the data in %

- Wilting point

Enter the data in %

- Method used for saturation, field capacity, wilting point

Report the type of methodology used according to table 26.

Table 26.

Method used for electric conductivity	Code
Pressure Plate Method	PP

Other	OTH
Unknown	UN

- Soil acidity: Exchangeable Al

Enter the data in meq/100g

- Method used for Soil acidity: Exchangeable Al

Report the type of methodology used

- pH (H_2O)

Enter the data

- Method used for pH (H_2O)

Report the type of methodology used according to table 27.

Table 27.

Method used for pH	Code
Soil water or salt solution ratio 1:1	OO
Soil water or salt solution ratio 1:2.5	OP
Other	OTH
Unknown	UN

- pH (KCl)

Enter the data

- Method used for pH (KCl)

Report the type of methodology used according to table 25.

- pH ($CaCl_2$)

Enter the data

- Method used for pH ($CaCl_2$)

Report the type of methodology used according to table 25.

- Organic Carbon content

Enter the data in g/Kg

- Method used for Organic Carbon content

Report the type of methodology used according to table 28.

Table 28

Method used for Organic Carbon Content	Code
Walkley –Black	WB
Elemental analyzer	EA
Springer & Klee	SK
Other	OTH
Unknown	UN

- Organic matter content

Enter the data in %

- Method used for organic matter content

Report the type of methodology used according to table 29.

Table 29.

Method used for Organic Matter Content	Code
Muffle	MF
Other	OTH
Unknown	UN

- $CaCO_3$ content

Enter the data in %

- Method used $CaCO_3$ content

Report the type of methodology used according to table 30.

Table 30.

Method used $CaCO_3$ content	Code
Calcimeter - Gas volumetric method	GV
Carbonate equivalent, Gravimetric method	GM
Titration method	TM
Other	OTH
Unknown	UN

- Gypsum content

Enter the data in %

- Method used for Gypsum content

Report the type of methodology used according to table 31.

Table 31.

Method used for Gypsum content	Code
Acetone, EDTA titration (George Holmgren)	AT
Soluble Calcium, FAAS	FA
Other	OTH
Unknown	UN

- CEC

Enter the data in cmol/Kg

- Method used for CEC

Report the type of methodology used according to table 32.

Table 32.

Method used for CEC	Code
$BaCl_2/MgSO_4$ method	BM
Ammonium Acetate (CH_3COONH_4)	AA
Mehlich Extraction	ME
Other	OTH
Unknown	UN

- Ca^{++}

Enter the data in cmol/Kg

- Method used for Ca^{++}

Report the type of methodology used according to table 33.

Table 33.

Method used for Ca^{++}	Code
Ammonium Acetate (CH_3COONH_4)	AA
Mehlich Extraction	ME
Other	OTH
Unknown	UN

- Mg^{++}

Enter the data in cmol/Kg

- Method used for Mg^{++}

Report the type of methodology used according to table 34.

Table 34.

Method used for Mg^{++}	Code
Ammonium Acetate (CH_3COONH_4)	AA
Mehlich Extraction	ME
Other	OTH
Unknown	UN

- Na^+

Enter the data in cmol/Kg

- Method used for Na^+

Report the type of methodology used according to table 35.

Table 35.

Method used for Na^+	Code
Ammonium Acetate (CH_3COONH_4)	AA
Mehlich Extraction	ME
Other	OTH
Unknown	UN

- K^+

Enter the data in cmol/Kg

- Method used for K^+

Report the type of methodology used according to table 36.

Table 36.

Method used for K^+	Code
Ammonium Acetate (CH_3COONH_4)	AA
Mehlich Extraction	ME
Other	OTH
Unknown	UN

- N tot content

Enter the data in g/Kg

- Method used for N tot content

Report the type of methodology used according to table 37.

Table 37.

Method used for N tot Content	Code
Kjeldahl	KJ
Elemental analyzer	EA
Coleman nitrogen analyzer	CA
Other	OTH
Unknown	UN

- Available P content

Enter the data in mg/kg

- Method used for available P content

Report the type of methodology used according to table 38.

Table 38.

Method used for available P Content	Code
Olsen	OL
Bray	BR

Mehlich No. 3	MN
Other	OTH
Unknown	UN

- Fe_{ox}

Enter the data in g/kg

- Fe_{d}

Enter the data in g/kg

- Fe_{p}

Enter the data in g/kg

- Fe tot

Enter the data in g/kg

- Mn

Enter the data in mg/kg

- Method used for Mn

Report the type of methodology used according to table 39.

Table 39.

Method used for Mn	Code
EDTA, FAAS	EF
DTPA, FAAS	DF
Other	OTH
Unknown	UN

- Zn

Enter the data in mg/kg

- Method used for Zn

Report the type of methodology used according to table 40.

Table 40.

Method used for Zn	Code
EDTA, FAAS	EF
DTPA, FAAS	DF
Other	OTH
Unknown	UN

- Cu

Enter the data in mg/kg

- Method used for Cu

Report the type of methodology used according to table 41.

Table 41.

Method used for Cu	Code
EDTA, FAAS	EF
DTPA, FAAS	DF
Other	OTH
Unknown	UN

- Active $CaCO_3$ content

Enter the data in %

Sheet 4. "Soil classification"

General information

This sheet has been inserted to collect the soil classification info of the profiles in the database.

List of new fields inserted

- Profile code

Enter a profile identification code, different for each profile.
The code must contain both letters and numbers. In particular, the code must consist of three letters and three numbers (e.g., AND001).
→ use the same code inserted in the field "Profile code" of sheet 1, 2 and 3.

- Horizons sequence

Report the horizons sequence of the studied profile.
e.g., A_B_Bt_C (horizon name, underscore, horizon name, underscore – without space)

- Old classification

Report the original classification of the soil profile.

- Classification system

Report the original classification system used to classify the soil according to Table 42.
Table 42. Classification system.

Classification system used	Code	Year	Code
USDA	US	1960	US60
		1975	US75
		1990	US90
		1991	US91
		1994	US94
		1998	US98
		1999	US99
		2006	US06
		2010	US10
		2014	US14
		2022	US22
FAO	FA	1967	FA67
		1974	FA74
		1985	FA85
		1988	FA88
		1994	FA94
WRB	WR	1998	WR98
		2006	WR06
		2015	WR15
		2022	WR22
CPCS	CP		
Referentiel Pedologique	RP	1995	RP95
		2008	RP08
KUBIENA	KU		
OTHER	OTH		
UNKNOWN	UN		

- New classification (WRB 4TH EDITION)

Report the new soil name based on the WRB 4TH EDITION.

The translation from the different reference systems used to WRB 4TH EDITION will be done on the basis of conversion tables, which will be provided later.